Fiscal Year:	FY 2025	**Task Last Updated:**	FY 11/01/2024
PI Name:	Fornace, Albert M.D.		
Project Title:	Space Radiation-Induced Persistent Estrogenic Response and Risk of Breast Cancer Development		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	http://www9.georgetown.edu/		
Project Type:	Ground	**Solicitation / Funding Source:**	2018 HERO 80JSC018N0001-Crew Health and Performance (FLAGSHIP, OMNIBUS). Appendix A-Flagship, Appendix B-Omnibus
Start Date:	01/02/2020	**End Date:**	05/01/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 05/01/2025 per S. Mack-Phillips/NASA-JSC (Ed., 5/8/24).		
Key Personnel Changes/Previous PI:	Ed. note (December 2020): The original Principal Investigator (PI) was Kamal Datta, M.D., who is now affiliated with NIH, as of early 2019. PI changed to Albert Fornace, M.D., before grant was awarded. In January 2023, the Co-Investigator (Dr. Michael Johnson) was replaced with Dr. Marc E. Lippman, who is a world-renowned breast cancer expert and Professor of Oncology at the Lombardi Comprehensive Cancer Center at Georgetown University.		
COI Name (Institution):	Brenner, David Ph.D. (Columbia University) Li, Xin Ph.D. (Georgetown University) Suman, Shubhankar Ph.D. (Georgetown University) Lippman, Marc (Georgetown University)		
Grant/Contract No.:	80NSSC19K1649		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Based on epidemiological analysis of radiation-exposed human cohorts, it is projected that space radiation could significantly increase breast cancer incidence and mortality among female astronauts. However, there is a large uncertainty in breast cancer risk estimation due to a lack of in vivo data. The proposed study aims to acquire in vivo quantitative and qualitative data on space radiation-induced breast cancer risk using a mouse model approach. Further, we also aim to test “persistent estrogenic response” (PER) signaling as a target to develop a potential countermeasure against space radiation-induced mammary tumorigenesis. We further hypothesize that targeting GCR-induced PER using an FDA-approved selective estrogen receptor modulator (SERM) will decrease the risk of mammary tumorigenesis. We plan to test our hypothesis by pursuing the following specific aims: Aim 1. Quantitatively compare effects of simulated GCR dose rates on mammary tumor incidence and grade in the ApcMin/+ tumor model. Aim 2. Characterize GCR dose-rate effects on PER in relation to mammary tumorigenesis. Aim 3. Determine roles of SERM in countering space radiation-induced PER and mammary tumorigenesis. Aim 4. Risk assessment (modeling) of mammary tumorigenesis after space radiation exposure. Key deliverables for this project are: 1) Develop a mathematical model for GCR dose rate-based breast cancer risk estimation, 2) Identify early markers of GCR-induced mammary tumorigenesis, and 3) Test potential mechanism-based FDA-approved countermeasure agents.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Breast cancer is the second most common cancer among women in the US. According to current breast cancer incidence data, approximately 12.9 percent of women in the United States will be diagnosed with breast cancer at some point in their lives. The annual rate of new cases and mortality from breast cancer in the United States is 128.5 per 100,000 and 20.1 per 100,000 women, respectively. Considering the high frequency of breast cancer in the American population, even a small increase by space radiation could have a major impact on risk estimates and planning of future space missions. In addition to risk estimation, studies on persistent estrogenic response after space radiation and its role in driving mammary tumorigenesis may provide insight into signaling events affecting carcinogenesis. The significance and deliverables of this project are to improve the estimates of breast cancer risk in women astronauts and to identify and test plausible targets for the development of mitigation strategies.
Task Progress:	The risk of space radiation-induced breast cancer in female astronauts is expected to increase during and after undertaking prolonged space missions, such as missions to Mars. However, studies comparing the effects of low and high dose rate space radiation exposures are few, and no studies are available in the literature comparing mammary tumorigenesis after low and high dose rate galactic cosmic radiation (GCR) exposures. Previous studies using mathematical and statistical model systems have demonstrated a significant probability of developing cancer after a Mars mission secondary to chronic radiation exposure. Despite these findings, large uncertainties still exist when making risk projections, mainly because there are limited epidemiological and biological data to describe the effects of GCR in human tissues. One general approach for risk assessment is to determine the relative biological effectiveness (RBE) of various parameters for space radiation compared to terrestrial radiation exposures. Since there is sufficient statistical sampling for the latter, risk estimates can then be “extrapolated” to space radiation using the RBE scaling factor (typically relative to gamma rays); needless to say, the reliability of the models for RBE determination is key. While the application of scaling factors is generally accepted to be the only practical approach to human cancer risk estimation for space radiation, a central, testable hypothesis is that qualitative and quantitative differences between space radiation and gamma-ray effects are maintained across species, such as from mouse to man. Understanding how to scale such risks in model systems will provide the best possible framework for undertaking the same scaling of cancer risks in humans. To use this approach requires the collection of relevant quantitative and qualitative data for oncogenic and pre-oncogenic endpoints in animal model systems relevant to human mammary cancer, as well as a sufficient understanding of the comparative molecular mechanisms involved in tumorigenesis. We proposed to use female ApcMin/+ mice to acquire the quantitative and qualitative biological data, using a range of radiation doses and dose rates simulating GCR, to develop a reliable risk prediction model that includes “both direct and non-targeted effects”. Additionally, dissecting the mechanisms of space radiation-induced persistent estrogenic response (PER) will allow evaluation of early markers for relevance as surrogates for late disease outcomes. Overall, the proposed research will increase our understanding of breast cancer risk from space radiation, and enhance NASA’s breast cancer risk prediction as well as countermeasure capabilities. NASA Task Book accomplishments (January to December 2024): [1] All required beam runs using acute and chronic 33-ion GCRsim were successfully completed in 2023, enabling comprehensive evaluation of samples from female ApcMin/+ and C57BL6 mice exposed to γ-rays and Galactic Cosmic Ray simulation (GCRsim) radiation (25–75 cGy) during January to December 2024. Compiled mammary tumorigenesis data in ApcMin/+ mice demonstrate a strong signal-to-noise ratio, with a clear dose-response observed for gamma, acute GCR, and chronic GCR exposures within the 25–75 cGy range. Both acute and chronic GCRsim exposures led to a higher rate of mammary tumor development compared to gamma exposure, with no apparent sparing effect from chronic versus acute GCRsim. Additional analyses on mammary carcinogenesis events, including established histological and molecular markers of preneoplastic lesions, revealed increased ductal outgrowth and a higher duct count in GCRsim-exposed ApcMin/+ mice relative to controls, accompanied by a significantly increased incidence of mammary tumors. [2] Analysis of GCRsim-exposed mice revealed a ~1.4-fold increase in serum estradiol concentrations at 100–110 days post-radiation compared to unirradiated controls, indicating systemic hormonal alterations. Immunohistochemical staining showed elevated protein expression and enhanced nuclear localization of ER and Cyclin D1 in mammary tissues of the GCRsim group. Quantitative analysis confirmed significant increases in ER and Cyclin D1 positive nuclei. Further gene expression analysis revealed significantly higher levels of ER downstream oncogenic targets, Ccnd1 and cMyc, suggesting that GCRsim exposure activates ER signaling, potentially promoting oncogenic pathways in mammary tissue. [3] GCRsim exposure significantly elevated Esrra mRNA levels and nuclear staining of ERR in mammary tissues, indicating activation of ERR through estrogen-related pathways. Analysis showed a marked increase in the expression of ERR downstream targets, Spp1 and Nrip1, in GCRsim-exposed mice compared to controls. Serum Spp1 protein levels were also significantly elevated (1.38-fold), with increased expression observed in mammary ductal epithelial cells. These results suggest that GCRsim irradiation enhances ERR signaling and its downstream pathways, potentially promoting tumorigenesis through estrogen-related mechanisms. [4] In mammary tumor tissues, GCRsim-exposed mice showed notable overexpression of ER, ERR, and Spp1

	compared to tumors in control mice, indicating a consistent upregulation of these protein markers in response to GCRsim exposure. This overexpression pattern suggests a potential link between GCRsim-induced signaling pathways and enhanced tumorigenic activity.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
Articles in Peer-reviewed Journals	Kumar K, Fornace AJ Jr, Suman S. "8-OxodG: A potential biomarker for chronic oxidative stress induced by high-LET radiation." DNA. 2024 Aug 1;4(3):221-38. https://doi.org/10.3390/dna4030015 ; PMID: 39268222; PMCID: PMC11391509 , Aug-2024
Papers from Meeting Proceedings	Kumar K, Angdisen J, Datta K, Fornace Jr AJ, Suman S. "Elevated SASP factors and reduced antioxidant enzymes in the serum of GCR-exposed mice coincide with increased mammary and gastrointestinal cancer incidence." 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. , Feb-2024
Papers from Meeting Proceedings	Kumar K, Angdisen J, Ma J, Datta K, Kallakury BVS, Fornace Jr AJ, Suman S. "Quantitative and qualitative assessment of galactic cosmic radiation-induced mammary tumorigenesis in ApcMin/+ mice. " 70th Annual Meeting of the Radiation Research Society (RRS), Tucson, Arizona, September 15-18, 2024. Abstracts. 70th Annual Meeting of the Radiation Research Society (RRS), Tucson, Arizona, September 15-18, 2024. , Sep-2024